Transfusion medicine

Products, thresholds and special processing · reactions · blood bank workup · refractoriness, massive hemorrhage, apheresis · **pages 1–2 on-call consult, page 3 board reference**

Compiled from the Fellowship Transfusion Medicine review, The Fellow on Call ep. 060, and AABB / ASFA / ISBT guidance · primary sources verified against PubMed, October 2026.

THE CONSULT

- **Transfuse for a physiologic reason, not a number.** Trend, symptoms, bleeding and comorbidity outweigh the absolute value; a stable Hb of 6.8 from iron deficiency may need IV iron, not blood
- **What is the deficit?** Hb, platelets, PT/INR + aPTT, **fibrinogen**, a specific factor
- **Bleeding or stable?** Thresholds below are for **non-bleeding** patients; active hemorrhage is treated to clinical endpoints
- **Blood bank history:** prior **alloantibodies**, prior reactions, **anti-CD38 therapy**, transplant (ABO switch), pregnancy
- **Special requirements:** irradiated, antigen-negative, HLA-matched, washed (table at right)
- **Goals / refusal:** ask product by product (many who decline RBCs accept albumin, factor concentrates, cell salvage); plan IV iron, ESA, TXA, minimal phlebotomy
- **Single-unit policy:** 1 RBC unit (or 1 apheresis platelet) → reassess before the next. Most over-transfusion is a reflexive 2-unit order
- **Stanford:** Transfusion Medicine pager **12027** · Apheresis pager **19164** (650-725-4656) · Apheresis fellow **27283** · Special Coag lab 650-723-4813

PRODUCTS

Product	Contents · adult dose → effect
RBCs	~300 mL, Hct ~55–65%. **1 unit → Hb ↑ ~1 g/dL** (Hct ~3%) in a non-bleeding adult
Platelets	1 apheresis unit ≈ a pool of 4–6 whole-blood-derived; **≥3×10^{11}**. **1 unit → ↑ ~30k** (less if refractory, febrile, splenomegalic). Half-dose = full dose for bleeding (PLADO)
Plasma (FFP / PF24)	All coagulation factors. **10–15 mL/kg** (~4 units in 70 kg) to correct a real coagulopathy
Cryo	**Fibrinogen**, FVIII, vWF, FXIII, fibronectin. **10 units (2 pre-pooled bags of 5) → fibrinogen ↑ ~70–100 mg/dL**
4F-PCC	II, VII, IX, X + proteins C/S. Warfarin + major bleed: **25 / 35 / 50 units/kg** for INR 2–<4 / 4–6 / >6 + **vitamin K 10 mg IV**

RBC THRESHOLDS — AABB 2023

Setting	Transfuse at Hb
Hospitalized, hemodynamically stable (incl. ICU)	**<7 g/dL** *strong rec.*
Heme/onc inpatients (chemo, HSCT)	**<7 g/dL** *conditional, low certainty*
Cardiac surgery	**<7.5 g/dL**
Orthopedic surgery or pre-existing CVD	**<8 g/dL** (FOCUS)
Septic shock	**<7 g/dL** (TRISS)
Acute MI + anemia	**Liberal <10 g/dL** *AABB 2025 conditional; ACC/AHA 2025 Class 2b* — MINT missed primary (RR 1.15, P=0.07); pooled meta-analysis: **cardiac death RR 1.47** with restrictive
Active bleeding / shock	Clinical endpoints, not a number

Fresh blood is not better: <8-day vs standard-issue RBCs did not change mortality (ABLE; RECESS) — don't request "fresh" units.

PLATELET THRESHOLDS — AABB/ICTMG 2025

Setting	Transfuse below
Chemo or allo-HSCT, not bleeding	**10k** *strong*
Auto-HSCT or aplastic anemia, not bleeding	**No routine prophylaxis** *conditional*
Consumptive (adult), no major bleeding	**10k**
Fever / sepsis	**~20k** *convention (Rebulla protocol)*
CVC, compressible site	**10k** *was 20k in 2015*
Lumbar puncture	**20k** *strong; was 50k in 2015*
IR procedure: low / high risk	**20k / 50k**
Major non-neuraxial surgery	**50k**
Neurosurgery, ocular, neuraxial	**~100k** *convention*
Active bleeding / CNS or massive bleed	**>50k / >100k** *convention*
ICH with plt >100k, incl. on antiplatelets	**Do not transfuse**

WHEN TO AVOID PLATELETS

- **TTP and HIT:** avoid unless life-threatening bleeding — fuels thrombosis (observational data)
- **ITP:** consumed within hours; only for serious bleeding, **with IVIG + steroids**
- **Post-transfusion purpura:** ineffective → **IVIG**
- Non-thrombocytopenic cardiac surgery without major bleeding

PLASMA · CRYO · REVERSAL

- **Plasma for:** bleeding + coagulopathy (liver, DIC, dilution), MTP, **TTP plasma-exchange replacement**, factor deficiency without a concentrate (FV)
- **Plasma misuse: don't correct INR <1.7 before a procedure** — it barely moves and adds TACO/TRALI risk; never as a volume expander; **not for warfarin reversal when PCC is available**
- **Cryo for fibrinogen <150 mg/dL with bleeding** (DIC, MTP, APL) — or fibrinogen concentrate
- **Dabigatran:** idarucizumab 5 g IV. **Xa inhibitors:** 4F-PCC (off-label) — **andexanet left the US market Dec 2025**

SPECIAL PROCESSING

Modification	Does · who needs it
Irradiated *25 Gy*	Inactivates donor T cells → **prevents TA-GVHD**. **Allo-HSCT** (from conditioning while immunosuppressed) · **auto-HSCT** (7 d before harvest → 3 mo post) · **Hodgkin lymphoma, any stage — indefinitely** · **purine analogs: fludarabine, cladribine, pentostatin, bendamustine — indefinitely** · **alemtuzumab** · congenital T-cell immunodeficiency · IUT/neonatal exchange · **HLA-matched or relative-donated units**, granulocytes · CAR-T per institutional policy. RBC shelf life drops to 28 d, K^+ rises
Pathogen-reduced plt	Amotosalen/UVA — **accepted alternative to irradiation**; also mitigates bacterial risk
Leukoreduced	<5×10^6 WBC/unit. ↓ FNHTR, ↓ HLA alloimmunization/refractoriness (TRAP), ↓ CMV. Most US units are pre-storage LR
CMV-safe	**LR ≈ CMV-seronegative** for HSCT, pregnancy, fetal/neonatal — either is acceptable
Washed	Removes plasma: **recurrent severe allergic reactions**, anaphylaxis with anti-IgA (or IgA-deficient donor), hyperkalemia risk
Antigen-negative	Known alloantibody; **sickle cell: Rh (C, E) + K matched**; anti-CD38 therapy (K-negative)
HLA-matched / crossmatched plt	**Immune refractoriness** (below). Irradiate

PLATELET REFRACTORINESS

- **CCI** = (post – pre count/μL) × BSA (m²) ÷ platelets given (×10^{11}). **1-h CCI <5,000–7,500 on 2 consecutive ABO-compatible units** = refractory
- **Good 1-h, poor 24-h → non-immune** (most common): fever, sepsis, DIC, splenomegaly, bleeding, drugs (amphotericin), VOD/TA-TMA
- **Poor 1-h → immune:** mostly **HLA class I** (HPA less often) → send **HLA class I antibody screen → HLA-matched, antigen-avoidant, or crossmatch-compatible** platelets, ABO-identical, irradiated
- Also think drug-induced immune thrombocytopenia, HIT, PTP. Prevention = leukoreduction (TRAP: 13% → 3–5%)
- Bleeding while refractory: keep transfusing + **antifibrinolytic** (TXA)

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ACUTE REACTION — FIRST 15 MINUTES

Any new sign during or after a transfusion

↓

1. STOP + support
stop the unit; keep the line open with NS
vitals, airway, oxygen; epinephrine if anaphylaxis
re-check patient ID vs unit label at bedside

↓

2. Notify blood bank (pager 12027)
return bag + tubing; repeat type and screen
DAT, CBC, LDH, bili, haptoglobin, free Hb, UA
blood cultures (patient + bag) if fever/rigors
or hypotension; CXR + BNP if dyspneic

↓

3. Classify by the dominant sign
fever alone → FNHTR, **after** excluding AHTR/sepsis
hives alone → allergic; may resume once resolved
hypotension → AHTR, sepsis, anaphylaxis, TRALI
hypoxemia → **TRALI vs TACO** (table below)

MASSIVE HEMORRHAGE

- **Activate per institutional MTP**; send T&S, CBC, PT/aPTT, fibrinogen, ionized Ca, ABG, lactate
- **1:1:1 plasma : platelets : RBC** (PROPPR vs 1:1:2: 24-h and 30-d mortality **not different**, but **hemostasis 86% vs 78%** and **fewer exsanguination deaths at 24 h, 9.2% vs 14.6%**)
- **TXA 1 g over 10 min + 1 g over 8 h, within 3 h** of trauma (CRASH-2: mortality RR 0.91). **After 3 h, bleeding deaths rose (RR 1.44) — don't start late**
- **Calcium** (citrate): follow iCa, replace; prevent **hypothermia** (warmer), acidosis; watch K^+
- **Fibrinogen <150 → cryo**; viscoelastic testing (TEG/ROTEM) where available; reverse anticoagulants

ORDERING AND THE BEDSIDE

- **Two-identifier bedside check** of patient vs unit — the single most important ABO safeguard
- **Only normal saline in the line** — LR's calcium clots citrated blood; hypotonic fluids hemolyze; no co-infused drugs
- **Finish each unit within 4 h** of issue (bacterial growth). Typical RBC unit over 1–2 h; **~1 mL/kg/h** if TACO-prone (HF, CKD, elderly, positive balance)
- **No routine premedication** — acetaminophen/diphenhydramine did not reduce reactions in randomized trials; reserve for recurrent reactors
- Consent covers reactions, TACO/TRALI and (very low) infection risk; document alternatives offered

REACTIONS

Reaction	Timing · hallmark	Management · prevention
AHTR	Minutes. **ABO mismatch (clerical)** → intravascular hemolysis: fever, flank pain, hemoglobinuria, shock, DIC, AKI	Stop; fluids for urine output, pressors, treat DIC. **Two-identifier bedside check**
FNHTR	During/<4 h; fever ± rigors; cytokines/anti-WBC	Antipyretic; diagnosis of exclusion. LR prevents
Allergic	Urticaria, pruritus only	Pause, antihistamine, resume. No routine premed
Anaphylaxis	Seconds–minutes; hypotension, bronchospasm. **Anti-IgA in IgA deficiency** (or haptoglobin deficiency)	Epinephrine. Check IgA; then **washed or IgA-deficient** products
Septic	High fever, rigors, shock; **platelets (room temp)** highest risk	Cultures (patient + bag), broad antibiotics; tell blood bank to quarantine co-components
Hypotensive	Isolated ↓BP within minutes; **ACE inhibitors** (bradykinin)	Resolves on stopping; hold ACE-i
DHTR	3–14 d. Anamnestic alloantibody (**Kidd** classic) → Hb drop, ↑bili, **DAT+ (IgG)**, new antibody	Usually supportive; antigen-negative units going forward
Hyperhemolysis	SCD: **Hb falls below pre-transfusion**; own + donor cells lysed, often DAT–	**Stop transfusing**; IVIG + steroids ± eculizumab
TA-GVHD	1–6 wk. Fever, rash, diarrhea, LFTs, **marrow aplasia**; **>90% fatal**	Prevention only: **irradiate or pathogen-reduce**
PTP	~5–10 d. Severe thrombocytopenia; **anti-HPA-1a**, classically multiparous women	**IVIG**; platelets ineffective; HPA-1a– if needed
Iron overload	Chronic transfusion (>~20 units) — heart, liver, endocrine	Ferritin, cardiac/liver T2* MRI; chelate

APHERESIS — ASFA 2023 (PAGER 19164)

Indication	Procedure · category
TTP	**Daily TPE — Cat I**; start on suspicion (PLASMIC), don't wait for ADAMTS13. + steroids + **caplacizumab** (HERCULES: recurrence 12% vs 38%) ± rituximab
Hyperviscosity (symptomatic)	TPE — **Cat I** (Waldenström, myeloma)
Leukostasis	Leukocytapheresis — **Cat III, not routine**; prophylactic also III. Cytoreduce (hydroxyurea, induction). **Avoid in APL**
Sickle cell	RBC exchange: **acute stroke Cat I**, stroke prophylaxis I, **acute chest II**. Target **HbS <30%**; avoid Hb >10–11
Other Cat I TPE	GBS, myasthenia gravis, CAPS, anti-GBM, ABO-incompatible / HLA desensitization (living-donor kidney)

BLOOD BANK WORKUP

- **Type and screen** = ABO/RhD + **antibody screen** (patient plasma vs reagent O cells, indirect antiglobulin test). Order it early in anyone likely to bleed
- **Crossmatch: electronic** (screen negative + 2 concordant ABO types) → minutes. **Full AHG crossmatch** if an alloantibody is present
- **Positive screen → antibody ID panel → antigen-negative units** — hours to days for multiple/rare antibodies. **Call early** in SCD, thalassemia, multiparous, chronically transfused
- **DAT** (patient RBCs): **IgG** ± C3 → warm AIHA, DHTR, drug; **C3 only** → cold agglutinin, PCH
- **Warm AIHA:** panreactive autoantibody; blood bank adsorbs to exclude hidden alloantibodies, then issues **"least incompatible" units — never withhold blood in life-threatening anemia**
- **Daratumumab** / **isatuximab** (CD38 on reagent RBCs) → **panreactive screen**. Before first dose: **type, screen + phenotype/genotype**, tell the blood bank. Lab uses **DTT-treated cells** → DTT destroys Kell, so give **K-negative units**
- **Emergency release:** uncrossmatched **O-neg** for females of childbearing potential and unknowns; **O-pos** for men and older women; **AB plasma**. Switch to type-specific ASAP
- **RhD-neg female of childbearing potential given RhD-pos platelets → RhIG** (residual RBCs can immunize)
- **ABO-mismatched HSCT:** blood bank picks the product group by transplant phase — don't override

TRALI VS TACO

	TRALI	TACO
Definition	**Within 6 h**; ARDS-like, no LAH. Type I (no ARDS risk factor) / II *Vlaar 2019*	**Within 12 h**; respiratory distress or pulmonary edema + ≥3 criteria *ISBT 2018*
Mechanism	Donor anti-HLA / anti-HNA → capillary leak	Hydrostatic volume overload
BP / JVP	**Hypotension**; JVP normal	**Hypertension**; JVP ↑
Fever	Often	Usually absent
BNP	Normal	↑ (NT-proBNP supports)
Diuretic	**No response — don't**	**Improves**
Prevent	Male/never-pregnant plasma donors	Slow rate, single units, pre-emptive diuretic in at-risk

Report both to the blood bank. TRALI does not require antibody detection to diagnose.

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BLOOD GROUPS AND ANTIBODIES

System	Antibody · clinical significance	Board pearl
ABO	Naturally occurring **IgM** anti-A/anti-B (group O also IgG) → **intravascular AHTR**	O = universal RBC donor; **AB = universal plasma donor**; plasma compatibility runs in reverse
Rh (D, C, c, E, e)	IgG; **D most immunogenic**. DHTR, **severe HDFN**	RhIG prevents anti-D only; passive anti-D makes the screen positive
Kell (K)	IgG; DHTR, **HDFN via erythroid suppression** (anemia > bilirubin)	~91% of donors K-negative; McLeod (absent Kx) → acanthocytes
Duffy (Fy^a/Fy^b)	IgG; DHTR, HDFN	**Fy(a–b–)** common with African ancestry → **P. vivax resistance** + **Duffy-null associated neutrophil count** (benign low ANC)
Kidd (Jk^a/Jk^b)	IgG, complement-fixing; **titers evanesce** below detection	**Classic cause of DHTR** — screen negative, then anamnestic boost
MNS	Anti-M/N usually cold IgM (insignificant); **anti-S, s, U are significant**	U-negative units are rare (African ancestry)
Lewis	IgM, insignificant; antigens adsorbed from plasma	Common in pregnancy; **no HDFN**
P	Anti-P1 insignificant. **Donath–Landsteiner**: biphasic IgG anti-P	**PCH** — children after viral illness; DAT C3+
I / i	Cold IgM autoantibodies	**Anti-I: Mycoplasma**, primary CAD · **anti-i: EBV**, lymphoma

Clinically significant = IgG reacting at 37°C/AHG (Rh, Kell, Duffy, Kidd, S/s). **Usually insignificant = IgM cold** (Lewis, M, N, P1, I). Mnemonic: "Kidd kills, Duffy dies, Lewis lives."

DAT PATTERNS

Result	Think
IgG ± C3	Warm AIHA (CLL, SLE, drugs), **DHTR** (mixed-field: only donor cells coated), HDFN, drug (penicillin, cephalosporins)
C3 only	**Cold agglutinin disease** (IgM, Mycoplasma, LPL/Waldenström), **PCH**, immune-complex drug
Negative + hemolysis	Spherocytosis, G6PD, MAHA, **PNH**; DAT-negative AIHA (low-affinity IgG, IgA)
Positive, no hemolysis	IVIG, ATG, recent RhIG; ~1 in several thousand healthy donors

ABO COMPATIBILITY

Recipient	RBCs	Plasma / cryo
O	O only	**O, A, B, AB**
A	A, O	A, AB
B	B, O	B, AB
AB	**A, B, AB, O**	AB only

RBCs: donor cells must lack the antigen the recipient's antibody sees. Plasma: the reverse. **Platelets** are given ABO-compatible when possible (out-of-group plasma can hemolyze; **ABO-identical improves increments** in refractoriness).

HDFN AND RH IMMUNE GLOBULIN

- **RhD-neg mother:** RhIG **300 µg at 28 wk + within 72 h of delivery** of an RhD-pos baby, and after bleeding/procedures
- **One 300-µg vial covers 30 mL fetal whole blood (15 mL fetal RBCs)**
- **Rosette screen → Kleihauer–Betke** (or flow) to size the bleed: fetal blood (mL) ≈ KB% × 50; vials = mL ÷ 30, round up + 1
- Once alloimmunized, RhIG is useless → titers, then **MCA Doppler**. **Anti-K: titers unreliable** — follow by Doppler
- IV anti-D for ITP needs RhD-pos, spleen intact; risk of intravascular hemolysis

STORAGE AND INFECTION RISK

Product	Storage · shelf life	Pearl
RBCs	1–6°C; **42 d** (additive solution); 28 d once irradiated	Storage lesion: K^+ leak, ↓2,3-DPG — age did not matter clinically (ABLE, RECESS)
Platelets	**20–24°C with agitation; 5 d** (7 d with FDA-approved bacterial testing)	**Highest bacterial risk** — FDA 2019 guidance: LVDS, secondary testing or **pathogen reduction**
Plasma	Frozen ≤–18°C, **1 yr**; thawed 24 h (thawed plasma 5 d at 1–6°C)	Labile FV/FVIII fall after thaw
Cryo	Frozen, **1 yr**; use within 4–6 h of thaw/pool	Room temperature after thaw

Viral residual risk in high-income systems with NAT: **HIV and HCV ~<1 per 1–2 million units**; HBV somewhat higher. **Non-infectious risks (TACO, TRALI, hemolysis) now dominate transfusion deaths.**

KEY TRIALS

Trial	Result → practice
TRICC 1999	ICU Hb 7 vs 10: 30-d mortality 18.7% vs 23.3% (P=0.11) → **restrictive is safe**
FOCUS 2011	Hip fracture + CV risk, Hb 10 vs symptoms/<8: death or inability to walk 35.2% vs 34.7% → restrictive OK in CVD
TRISS 2014	Septic shock, Hb 7 vs 9: 90-d mortality 43.0% vs 45.0% (RR 0.94) → 7 in sepsis
MINT 2023	Acute MI, liberal <10 vs restrictive 7–8 (n=3,504): MI/death 14.5% vs 16.9%, **RR 1.15 (0.99–1.34), P=0.07**; death 8.3% vs 9.9% → liberal favored, **not proven**
ABLE / RECESS 2015	Fresh (<8 d / ≤10 d) vs standard RBCs: no mortality or organ-dysfunction difference → **don't request fresh units**
Rebulla 1997	AML induction, plt threshold 10k vs 20k: major bleeding 21.5% vs 20%, **21.5% fewer platelet transfusions** → the 10k threshold
PLADO 2010	Platelet **dose** (1.1 / 2.2 / 4.4 $\times 10^{11}$/m²) at ≤10k: grade ≥2 bleeding 71 / 69 / 70% → **dose doesn't matter; a dose trial, not a threshold trial**
TOPPS 2013	No-prophylaxis vs prophylaxis <10k: bleeding 50% vs 43%, **non-inferiority not shown** → keep prophylaxis after chemo/allo
TRAP 1997	AML: leukoreduced or UVB platelets cut alloimmune refractoriness 13% → 3–5%
PROPPR 2015	Trauma 1:1:1 vs 1:1:2: 30-d mortality 22.4% vs 26.1% (P=0.26); hemostasis 86% vs 78%; exsanguination death 9.2% vs 14.6%
CRASH-2 2010/11	Trauma TXA: all-cause death 14.5% vs 16.0% (RR 0.91); benefit only if given ≤3 h
HERCULES 2019	iTTP caplacizumab + TPE: composite TTP death/recurrence/TE 12% vs 49%; recurrence 12% vs 38%

SICKLE CELL AND CHRONIC TRANSFUSION

- **Prophylactic Rh (C, E) + K antigen matching** for all SCD transfusions; extended matching or RH genotyping if alloimmunized
- **Simple transfusion** for symptomatic anemia; **exchange** to lower HbS (stroke, severe ACS, high-risk surgery). **Never push Hb >10–11** — viscosity
- **DHTR in SCD mimics a pain crisis** — check HbA% and a new antibody; hyperhemolysis = stop transfusing
- **Iron overload:** chelate with deferasirox, deferoxamine or deferiprone (agranulocytosis — weekly ANC)